

- 17** A single-celled organism lives in freshwater. Water that enters the cytoplasm of the cell by osmosis is collected into a structure called the contractile vacuole. To remove the water the contractile vacuole fuses with the cell surface membrane.

A student counted the number of times that the contractile vacuole filled and emptied when the cell was placed in solutions with different water potentials. The results are shown in the table.

water potential of external solution / kPa	rate of contractile vacuole emptying / min ⁻¹
0	31
-100	20
-200	13
-300	8
-400	6
-500	0

Which statement explains the pattern observed as the water potential of the external solution decreased?

- A** The water potential gradient between the cell and the solution increased, causing water to move into the cell more rapidly and the contractile vacuole to empty more frequently.
- B** The water potential gradient between the cell and the solution increased, causing water to move into the cell less rapidly and the contractile vacuole to empty less frequently.
- C** The water potential gradient between the cell and the solution decreased, causing water to move into the cell more rapidly and the contractile vacuole to empty more frequently.
- D** The water potential gradient between the cell and the solution decreased, causing water to move into the cell less rapidly and the contractile vacuole to empty less frequently.